Package ‘rENM.model’

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Type Package

Title Modeling tools for the rENM framework

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Description The rENM framework is a modular system for reconstructing and analyzing long-term ecological niche dynamics using time-indexed environmental data and species occurrence records. Its packages support climate-based niche modeling, temporal analysis of climatic suitability, and reproducible workflows for studying ecological change.

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URL https://github.com/rENM-Framework/rENM.model

BugReports https://github.com/rENM-Framework/rENM.model/issues

Imports rENM.core,
dismo,
doParallel,
dplyr,
foreach,
ggplot2,
maxnet,
raster,
scales,
sdm,
sf,
sp,
terra,
usdm

Contents

`create_ensemble_model` *Build an ensemble ENM and derived products for a species*

Description

Constructs an ensemble ecological niche model (ENM) for a given species (four-letter banding code) and a 5-year time slice using the sdm framework, producing ensemble predictions and derived products including range maps and model diagnostics.

Usage

```
create_ensemble_model(
  alpha_code,
  year,
  methods = c("maxnet", "rf", "brt", "glm", "mars"),
  reps = 3,
  bg = 2500,
  tp = 40,
  op = 2,
  ensemble_method = "unweighted",
  io = c("raster", "terra"),
  overwrite = TRUE,
  verbose = TRUE
)
```

Arguments

`alpha_code`	Character. Four-letter banding code, case-insensitive (validated as exactly four A-Z letters; stored upper-case).
`year`	Integer. A 5-year bin start in (1980, 2020) divisible by 5.
`methods`	Character. SDM algorithms to fit. Defaults to `c("maxnet","rf","brt","glm","mars")`.
`reps`	Integer. Number of replicated subsampling runs. Default `3`.
`bg`	Integer. Number of background points. Default `2500`.
`tp`	Numeric. Test partition percent (0-100). Default `40`.
`op`	Integer. Threshold optimization option passed to `sdm::getEvaluation` for binarization; `2` = max(se+sp). Default `2`.

`ensemble_method`
: Character. Ensemble combination method passed to `sdm::ensemble` in `setting = list(method = ...)`. Default `"unweighted"`.

`io`
: Character. Raster I/O backend for writing outputs; one of `"raster"` or `"terra"`. Predictors are read via raster either way to maintain sdm compatibility. Default `"raster"`.

`overwrite`
: Logical. Overwrite existing output files. Default `TRUE`.

`verbose`
: Logical. Emit timestamped console progress. Default `TRUE`.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context: Inputs/paths must follow the rENM run structure:

```
<rENM_project_dir()>/runs/<ALPHA>/TimeSeries/<YEAR>/occs/of-<YEAR>.csv
<rENM_project_dir()>/runs/<ALPHA>/TimeSeries/<YEAR>/vars/*.asc
<rENM_project_dir()>/runs/<ALPHA>/TimeSeries/<YEAR>/model/  (outputs written here)
```

Inputs: The occurrence CSV must include `longitude`, `latitude`.

Methods: Models are trained via `sdm::sdm()` with replicated subsampling (`reps`) and a test partition (`tp`, percent). Background points (bg) are drawn using `method = "gRandom"`.

The default modeling ensemble uses `"maxnet"` (a native R implementation of maximum entropy modeling) instead of `"maxent"` to avoid Java dependencies and improve reproducibility in CRAN-compliant environments.

Outputs:

- Ensemble prediction rasters (GeoTIFF and ASCII)
- Model statistics and variable importance summaries
- Binary range map derived via threshold optimization

Logging: A human-readable, append-only summary is written to: `<rENM_project_dir()>/runs/<alpha_code>/_log.txt` with key parameters, file outputs, and per-year status. Console messages include timestamps for traceability.

Value

Invisibly returns a List with structured elements:

- `data`: sdmData object containing training data
- `model`: sdmModel object representing fitted models
- `ensemble`: RasterLayer of predicted suitability
- `range`: RasterLayer of binary presence/absence
- `paths`: List of important written file paths
- `stats`: List of captured model statistics and variable importance outputs

Side effects include writing raster outputs, statistics files, variable importance summaries, plots, and appending a log entry to the run log file.

See Also

sdm, sdmData, ensemble, getEvaluation

Examples

```
## Not run:
# Default knobs; raster I/O
result <- create_ensemble_model("CASP", 2000)

# Heavier replication, custom methods, terra I/O
result <- create_ensemble_model(
  alpha_code = "CASP", year = 2000,
  methods = c("maxnet","glm","brt","rf","mars"),
  reps = 5, bg = 5000, tp = 30, op = 2,
  ensemble_method = "unweighted",
  io = "terra", overwrite = TRUE, verbose = TRUE
)

## End(Not run)
```

`create_range_map` *Create and save a binary presence-absence range map*

Description

Build a ggplot2 map for a binary (0/1) presence-absence raster, overlaid with state boundaries and the species GAP range polygon. Writes PNG, ASC, and GeoTIFF outputs to the per-year model directory.

Usage

```
create_range_map(
  pa,
  alpha_code,
  year,
  state_shp = NULL,
  title = "Range Map",
  layer = 1L,
  minor_div = 2L,
  width_px = 1000,
  height_px = 750,
  dpi = 72,
  title_size = 25,
  legend_size = 15,
  legend_title_size = 15,
  axis_title_size = 20,
  axis_text_size = 18
)
```

Arguments

pa	SpatRaster, Character, or Raster. A presence-absence raster.
alpha_code	Character. Species code (e.g., "CASP"), used to resolve output paths and to look up GAP.RANGE via get_species_info().
year	Integer or Character. Year inserted into the output path and name.
state_shp	Character. Path to a states/administrative-boundaries shapefile. Default resolves to <project_dir>/data/shapefiles/tl_2012_us_state/tl_2012_us_state.shp.
title	Character. Plot title. Default "Range Map".
layer	Integer. Layer index when pa has multiple layers. Default 1L.
minor_div	Integer. Minor grid divisions between major axis breaks. Default 2L.
width_px, height_px	Numeric. PNG dimensions in pixels. Defaults 1000 and 750.
dpi	Numeric. Plot DPI for ggsave(). Default 72.
title_size, legend_size, legend_title_size, axis_title_size, axis_text_size	Numeric. Font sizes for plot elements.

Details

Output files are written to: <project_dir>/runs/<alpha_code>/TimeSeries/<year>/model/

The species range shapefile is resolved via get_species_info(alpha_code), which looks up GAP.RANGE in the species table and reads the shapefile from <project_dir>/data/shapefiles/<gap_range>/<gap_ra
State and range vectors are reprojected to the raster CRS if necessary. Values in the raster are coerced to binary by round(). Sidecar files (.prj, .aux.xml) are removed after writing.

Value

Invisible list with plot (ggplot2 object), img_path, asc_path, tif_path. Side effects are raster and image files written to disk and sidecar files removed.

See Also

plot_suitability, rENM_project_dir

Examples

```
## Not run:
  create_range_map("rasters/casp_1980_pa.tif", "casp", 1980)

  r <- terra::rast("rasters/casp_stack.tif")
  create_range_map(r, "casp", 1990, layer = 2L, title = "CASP 1990 Range")

## End(Not run)
```

create_timeseries *Build a full rENM time series for a species*

Description

Schedules and executes `create_ensemble_model` for each 5-year bin from 1980 to 2020 in parallel, producing a complete rENM time series for a given species.

Usage

```
create_timeseries(alpha_code, project_dir = NULL)
```

Arguments

`alpha_code`	Character. Four-letter banding code (e.g., "CASP").
`project_dir`	Character. Path to the rENM project root. If NULL, resolved via `rENM_project_dir`.

Details

Workers are launched in a PSOCK cluster of size `min(number_of_years, detectCores())`. Each year is processed independently so errors for individual years are captured and logged without aborting the full time series. The cluster is always stopped on exit.

`create_ensemble_model()` is referenced via namespace closure and is available to workers in both installed-package and `devtools::load_all()` workflows.

A processing summary is appended to `<project_dir>/runs/<alpha_code>/_log.txt`.

Value

Invisible named list keyed by year. Each element contains: `ok` (logical), `year` (integer), `elapsed` (seconds), `result` (output from `create_ensemble_model()` or NULL), `notes` (character: "ok" or an error message). The attribute `total_elapsed_sec` records total wall-clock time. Side effects are parallel job execution, file outputs per year, and a summary log entry.

See Also

`create_ensemble_model`, `rENM_project_dir`

Examples

```
## Not run:
  create_timeseries("CASP")
  create_timeseries("CASP", project_dir = "/my/project")

## End(Not run)
```

`plot_suitability` *Plot a climatic suitability raster*

Description

Build a ggplot2 map of continuous climatic suitability values from a raster, overlaid with state boundaries and the species GAP range polygon.

Usage

```
plot_suitability(
  en,
  alpha_code,
  state_shp = NULL,
  title = "Climatic Suitability Map",
  limits = c(0, 1),
  layer = 1L,
  minor_div = 2L
)
```

Arguments

`en`	SpatRaster, Character, or Raster. A suitability raster.
`alpha_code`	Character. Species code used to look up `GAP.RANGE` via `get_species_info()`.
`state_shp`	Character. Path to a states/administrative-boundaries shapefile. Default resolves to `<project_dir>/data/shapefiles/tl_2012_us_state/tl_2012_us_state.shp`.
`title`	Character. Plot title. Default `"Climatic Suitability Map"`.
`limits`	Numeric. Length-2 vector for the color scale limits; default `c(0, 1)`.
`layer`	Integer. Band index when `en` has multiple layers; default 1L.
`minor_div`	Integer. Minor grid divisions between major axis breaks; default 2L.

Details

The species range shapefile is resolved via `get_species_info(alpha_code)`, which looks up `GAP.RANGE` and reads the shapefile from `<project_dir>/data/shapefiles/<gap_range>/<gap_range>.shp`. State and range vectors are reprojected to the raster CRS if necessary. NA raster cells render transparent.

Value

A ggplot2 object. No files are written.

See Also

`save_suitability_plot`, `rENM_project_dir`

Examples

```
## Not run:
  p <- plot_suitability(en, "CASP")
  p <- plot_suitability(en, "CASP", title = "CASP 2000 Suitability")

## End(Not run)
```

---

rank_variable_importance

Parse and rank variable importance from an SDM text report

Description

Parse a "Relative Variable Importance" style text report and return a tidy data frame. Supports multiple metric sections and optional aggregation across metrics.

Usage

```
rank_variable_importance(
  input,
  combine_metrics = TRUE,
  log = FALSE,
  log_path = NULL,
  alpha_code = NULL
)
```

Arguments

input	Character. File path, single string, or vector of lines.
combine_metrics	Logical. Average across metrics if TRUE. Default TRUE.
log	Logical. Write summary to log. Default FALSE.
log_path	Character. Optional explicit log file path.
alpha_code	Character. Optional 4-letter species code used to resolve the default log path when `log_path` is not supplied.

Details

The parser expects one or more metric sections of the form:

```
Based on <Metric> metric:
var1 (40
var2 (20
====
```

Input forms:

- A path to a readable file (read line-by-line).
- A single character string (split on `\n`).

- A character vector (treated as pre-split lines).

Variable names may include letters, digits, underscores, hyphens, and dots. A permissive fallback regex is tried when the strict pattern fails. Percentages are not constrained to (0, 100).

When combine_metrics = TRUE, values are averaged across metrics (mean(..., na.rm = TRUE)) and sorted descending by percent.

When log = TRUE, a summary block is appended to <project_dir>/runs/<alpha_code>/_log.txt (if alpha_code is supplied) or to log_path.

Value

Data frame with columns variable and percent (combined) or additionally metric (uncombined), sorted by descending importance.

See Also

create_ensemble_model, rENM_project_dir

Examples

```
## Not run:
  df <- rank_variable_importance("file.txt")
  df <- rank_variable_importance(pi0, combine_metrics = TRUE)

## End(Not run)
```

---

reduce_covariance   *Reduce variable covariance via adaptive VIF screening*

Description

Detects multicollinearity in staged predictor rasters through an adaptive VIF routine, moves correlated predictors to a sub-directory, and writes a results CSV. Processes one or more 5-year time bins.

Usage

```
reduce_covariance(
  alpha_code,
  year = NULL,
  vif_threshold = 10,
  sample_schedule = c(10000, 20000, 50000),
  stability_patience = 1,
  corr_threshold = 0.95,
  exclude_vars = character(0),
  file_patterns = c("\\.tif$", "\\.grd$", "\\.img$", "\\.bil$"),
  overwrite_results = TRUE
)
```

Arguments

- `alpha_code` — Character. Species alpha code (e.g., "CASP").
- `year` — Integer vector or NULL. Years to process; NULL uses `seq(1980, 2020, by = 5)`.
- `vif_threshold` — Numeric. VIF threshold for `usdm::vifstep()`; default 10.
- `sample_schedule` — Integer vector. Adaptive sample sizes tried in order; stops early once the retained set stabilizes. Default `c(10000, 20000, 50000)`.
- `stability_patience` — Integer. Consecutive identical retained sets required for convergence; default 1.
- `corr_threshold` — Numeric. Absolute Pearson correlation threshold for reporting high-correlation pairs; default 0.95.
- `exclude_vars` — Character vector. Predictor names always retained regardless of VIF result; default `character(0)`.
- `file_patterns` — Character vector. Regex patterns selecting raster filenames to include. Default `c("\.tif$", "\.grd$", "\.img$", "\.bil$")`.
- `overwrite_results` — Logical. Overwrite an existing results CSV; default TRUE.

Details

Input rasters are read from `<project_dir>/runs/<alpha_code>/TimeSeries/<year>/vars/`. The function samples `sample_schedule` sizes in order, running `usdm::vifstep()` at each size until the retained set is stable for `stability_patience` consecutive iterations.

Outputs per year:

- Results CSV: `_<alpha_code>-<year>-Correlation-Results.csv`
- Moved rasters (only when variables are dropped): `_<alpha_code>-<year>-Correlated-Variables/`
- Log entry appended to `_log.txt`

Value

Invisible named list keyed by year. Each element contains: `kept`, `dropped`, `vif_table`, `cor_pairs`, `results_csv`, `moved_to` (directory path or NA).

See Also

`rENM_project_dir`, `stage_screened_variables`

Examples

```
## Not run:
  reduce_covariance("CASP")
  reduce_covariance("CASP", year = 2005, exclude_vars = c("bio1", "bio12"))

## End(Not run)
```

save_suitability_plot *Save a ggplot suitability plot to file*

Description

Save a ggplot object to disk with configurable dimensions, resolution, and typography. File format is inferred from the filename extension or defaults to PNG. Relative paths are resolved against the project directory.

Usage

```
save_suitability_plot(
  p,
  fn,
  width = 1000,
  height = 750,
  res = 72,
  pointsize = 15,
  title_size = 28,
  subtitle_size = 20,
  legend_size = 15,
  legend_title_size = 15,
  axis_title_size = 20,
  axis_text_size = 18
)
```

Arguments

`p`	ggplot. The ggplot object to save.
`fn`	Character. Output path with or without extension. If no extension is provided, `.png` is appended. Relative paths are resolved against the project directory.
`width`	Numeric. Image width in pixels. Default 1000.
`height`	Numeric. Image height in pixels. Default 750.
`res`	Numeric. Resolution in DPI. Default 72.
`pointsize`	Numeric. Base point size passed to `ggsave()`. Default 15.
`title_size`	Numeric. Font size for the plot title. Default 28.
`subtitle_size`	Numeric. Font size for the plot subtitle. Default 20.
`legend_size`	Numeric. Font size for legend text. Default 15.
`legend_title_size`	Numeric. Font size for the legend title. Default 15.
`axis_title_size`	Numeric. Font size for axis titles. Default 20.
`axis_text_size`	Numeric. Font size for axis tick labels. Default 18.

Details

Width and height are specified in pixels and converted to inches using `res` (DPI) for `ggplot2::ggsave()`. The plot theme is augmented with the specified font sizes before saving. Supported extensions: `png`, `pdf`, `tiff`, `jpg`, `jpeg`, `svg`, `eps`.

Value

Invisibly returns the file path written. Side effect: writes an image file to disk.

See Also

plot_suitability, ggsave

Examples

```
## Not run:
  p <- ggplot2::ggplot(mtcars, ggplot2::aes(mpg, wt)) + ggplot2::geom_point()
  save_suitability_plot(p, "plots/mtcars_scatter.png")

  # No extension -> defaults to PNG:
  save_suitability_plot(p, "plots/mtcars_scatter")

## End(Not run)
```

---

screen_by_convergence1 *Iterative bivariate ENM screening via dismo MaxEnt (convergence 1)*

Description

Repeatedly fits bivariate MaxEnt models (via **dismo**), aggregates permutation importance (PI) scores, and stops when both the mean PI and membership of the adaptive top-k variable set stabilize. Use this variant when Java-based MaxEnt is available; see screen_by_convergence2 for a Java-free alternative.

Usage

```
screen_by_convergence1(
  alpha_code,
  year = NULL,
  max_iter = 50,
  batch_samples = NULL,
  w_stability = 0.5,
  background_n = 2500,
  seed = NULL,
  ncores = max(1, parallel::detectCores() - 1),
  fast = FALSE,
  set_threshold = 0.9,
  passes_required = NULL
)
```

Arguments

alpha_code	Character. Species or run code (e.g., "CASP").
year	Integer, Character, or NULL. Year to process; NULL runs all 5-year intervals 1980–2020.

`max_iter` Integer. Maximum iterations safeguard; default 50.

`batch_samples` Integer or NULL. Target appearances per variable per iteration; NULL triggers an adaptive fallback.

`w_stability` Numeric. Instability weight in `wz_score`; default 0.5.

`background_n` Integer. Background points per model; default 2500.

`seed` Integer or NULL. RNG seed; NULL draws a random seed that is returned in the result.

`ncores` Integer. CPU cores for parallel fits; default `max(1, parallel::detectCores() - 1)`.

`fast` Logical. Enable fast triage mode; default FALSE.

`set_threshold` Numeric. Jaccard threshold for membership stability; default 0.90.

`passes_required` Integer or NULL. Consecutive passes needed for convergence; NULL defaults to 1 in fast mode, else 2.

Details

Reads occurrences from `runs/<alpha_code>/_occs/of-<year>.csv` and raster predictors from `runs/<alpha_code>/_vars/<year>/`. Writes a variable-ranking CSV and a convergence-trace CSV to `runs/<alpha_code>/TimeSeries/<year>/vars/`.

Convergence criteria (both must pass):

- *Mean stability* – relative change in top-k mean PI below an adaptive threshold (10% if $p \leq 20$, 7.5% if $21 \leq p \leq 40$, else 5%).
- *Membership stability* – Jaccard overlap of consecutive top-k sets above `set_threshold`.

Adaptive top-k: $k = \min(12, \max(5, \text{round}(\sqrt{p})))$. Batch sampling defaults to `max(1, floor(1.5 * sqrt(p)))` appearances per variable (halved in `fast` mode). Parallel execution uses deterministic per-worker RNG streams.

If `year = NULL`, all 5-year intervals 1980–2020 are processed in sequence.

Value

Invisible list with elements:

`importance_summary` Data frame of variable-wise PI statistics and rank diagnostics.

`convergence_trace` Data frame with per-iteration diagnostics.

`output_files` Named list with paths to the ranking and convergence-trace CSV files.

`seed_used` Integer seed used in this run.

Side effects: two CSV files and a log block are written to disk.

See Also

`screen_by_convergence2`, `rENM_project_dir`

Examples

```
## Not run:
## Quick triage across all years
res <- screen_by_convergence1("CASP", fast = TRUE)

## Standard fidelity for a single year
res <- screen_by_convergence1("CASP", 1990, seed = 42)

## End(Not run)
```

---

screen_by_convergence2

Iterative bivariate ENM screening via native maxnet (convergence 2)

Description

Repeatedly fits bivariate `maxnet` models, aggregates permutation importance (PI) scores, and stops when both the mean PI and membership of the adaptive top-k variable set stabilize. Identical convergence logic to `screen_by_convergence1` but uses **maxnet** instead of Java-based **dismo** MaxEnt, eliminating the Java dependency.

Usage

```
screen_by_convergence2(
  alpha_code,
  year = NULL,
  max_iter = 50,
  batch_samples = NULL,
  w_stability = 0.5,
  background_n = 2500,
  seed = NULL,
  ncores = max(1, parallel::detectCores() - 1),
  fast = FALSE,
  set_threshold = 0.9,
  passes_required = NULL
)
```

Arguments

`alpha_code`	Character. Species or run code (e.g., "CASP").
`year`	Integer, Character, or NULL. Year to process; NULL runs all 5-year intervals 1980–2020.
`max_iter`	Integer. Maximum iterations safeguard; default 50.
`batch_samples`	Integer or NULL. Target appearances per variable per iteration; NULL triggers an adaptive fallback.
`w_stability`	Numeric. Instability weight in `wz_score`; default 0.5.
`background_n`	Integer. Background points per model; default 2500.
`seed`	Integer or NULL. RNG seed; NULL draws a random seed that is returned in the result.

`ncores` Integer. CPU cores for parallel fits; default `max(1, parallel::detectCores() - 1)`.

`fast` Logical. Enable fast triage mode; default FALSE.

`set_threshold` Numeric. Jaccard threshold for membership stability; default 0.90.

`passes_required`
Integer or NULL. Consecutive passes needed for convergence; NULL defaults to 1 in fast mode, else 2.

Details

Reads occurrences from `runs/<alpha_code>/_occs/of-<year>.csv` and raster predictors from `runs/<alpha_code>/_vars/<year>/`. Writes a variable-ranking CSV and a convergence-trace CSV to `runs/<alpha_code>/TimeSeries/<year>/vars/`.

Background points are drawn directly from non-NA raster cells using `raster::sampleRandom(xy = TRUE, na.rm = TRUE)`. Permutation importance is computed as a MaxEnt-like AUC-drop metric: each predictor is permuted in turn, the drop in training AUC is measured without refitting, and drops are normalized to percentages summing to 100 within each bivariate model.

Convergence criteria (both must pass):

- *Mean stability* – relative change in top-k mean PI below an adaptive threshold (10% if $p \leq 20$, 7.5% if $21 \leq p \leq 40$, else 5%).
- *Membership stability* – Jaccard overlap of consecutive top-k sets above `set_threshold`.

Adaptive top-k: $k = \min(12, \max(5, \mathrm{round}(\sqrt{p})))$. Batch sampling defaults to `max(1, floor(1.5 * sqrt(p)))` appearances per variable (halved in `fast` mode). Parallel execution uses deterministic per-worker RNG streams. Balanced pairing prevents self-pairs.

If `year = NULL`, all 5-year intervals 1980–2020 are processed in sequence.

Value

Invisible list with elements:

`importance_summary` Data frame of variable-wise PI statistics and rank diagnostics.

`convergence_trace` Data frame with per-iteration diagnostics.

`output_files` Named list with paths to the ranking and convergence-trace CSV files.

`seed_used` Integer seed used in this run.

Side effects: two CSV files and a log block are written to disk.

See Also

`screen_by_convergence1`, `rENM_project_dir`

Examples

```
## Not run:
## Quick triage across all years
res <- screen_by_convergence2("CASP", fast = TRUE)

## Standard fidelity for a single year
res <- screen_by_convergence2("CASP", 1990, seed = 42)

## End(Not run)
```

stage_all_variables *Stage environmental variables into TimeSeries bins*

Description

Copies raster variables (`.tif`, `.asc`) from the run-level `_vars/<year>/` directory into `TimeSeries/<year>/vars/` for downstream modeling.

Usage

```
stage_all_variables(
  alpha_code,
  year = seq(1980, 2020, by = 5),
  project_dir = NULL
)
```

Arguments

alpha_code	Character. Species alpha code (e.g., "CASP").
year	Integer. Vector of 5-year bins. Default: `seq(1980, 2020, by = 5)`.
project_dir	Character. Path to the rENM project root. If NULL, resolved via `rENM_project_dir`.

Details

Files are copied flat (subdirectory structure is not preserved). Existing destination files are overwritten. Duplicate basenames within a year trigger a warning; the last file with that name wins. A processing summary is appended to the run log.

Value

Invisible data frame with columns `year`, `src_dir`, `dest_dir`, `files_found`, `files_copied`, `duplicates`. Primary side effects are file copies, directory creation, and a log entry appended to `_log.txt`.

See Also

`rENM_project_dir`, `stage_occurrences`

Examples

```
## Not run:
  stage_all_variables("CASP")
  stage_all_variables("CASP", year = c(1990, 2010, 2020))

## End(Not run)
```

`stage_occurrences` *Stage occurrence CSVs into TimeSeries bins*

Description

Copies per-bin occurrence files from the run-level `_occs` directory into `TimeSeries/<year>/occs/` for downstream modeling.

Usage

```
stage_occurrences(alpha_code, year = seq(1980, 2020, 5), project_dir = NULL)
```

Arguments

`alpha_code`	Character. Species alpha code (e.g., `"CASP"`).
`year`	Integer. Vector of 5-year bins in {1980, 1985, ..., 2020}. Default: `seq(1980, 2020, 5)`.
`project_dir`	Character. Path to the rENM project root. If NULL, resolved via `rENM_project_dir`.

Details

Source files (`of-<year>.csv`) are read from `<project_dir>/runs/<alpha_code>/_occs/` and written to `<project_dir>/runs/<alpha_code>/TimeSeries/<year>/occs/`. Existing destination files are overwritten. A formatted audit block is appended to the run log and echoed to the console.

Value

Invisible list with elements: `alpha_code`, `years`, `copied_paths`, `counts` (list: `total_years`, `valid_files`, `copied_files`, `missing_files`), `elapsed_seconds`, `log_file`. Primary side effects are file copies, directory creation, and a log entry appended to `_log.txt`.

See Also

`rENM_project_dir`, `stage_all_variables`

Examples

```
## Not run:
  stage_occurrences("CASP")
  stage_occurrences("CASP", year = c(1980, 2000, 2015))

## End(Not run)
```

stage_screened_variables

Stage screened environmental variables into TimeSeries bins

Description

Copies ranked raster predictor layers into per-year TimeSeries/<year>/vars/ directories based on a variable-ranking CSV produced by screen_by_convergence1 or screen_by_convergence2. Supports several ranking metrics and an optional top-N filter.

Usage

```
stage_screened_variables(
  alpha_code,
  year = seq(1980, 2020, 5),
  top_n = 10,
  rank_method = "ratio"
)
```

Arguments

alpha_code	Character. Species alpha code (e.g., "CASP").
year	Integer. Vector of 5-year bins in {1980, 1985, ..., 2020}. Default: seq(1980, 2020, 5).
top_n	Integer or NULL. Number of top-ranked variables to stage per year; NULL copies all variables. Default 10.
rank_method	Character. Ranking metric; one of "ranksum", "medianPI", "ratio", "wz". Default "ratio".

Details

Source rasters are read from <project_dir>/runs/<alpha_code>/_vars/<year>/. Rankings are read from <project_dir>/runs/<alpha_code>/TimeSeries/<year>/vars/ _<alpha_code>-<year>-Variable-Ra A processing summary is appended to the run log.

Supported rank_method values and the corresponding ranking column:

- "ratio" – rank_ratio (default)
- "ranksum" – rank_ranksum
- "medianPI" – rank_medianPI
- "wz" – rank_wz

Value

Invisible named list with elements: alpha_code, years, rank_method, top_n, copied_paths, counts (list: total_years, attempted_files, copied_files, missing_files, years_missing_ranking), elapsed_seconds, log_file. Primary side effects are file copies, directory creation, and a log entry appended to _log.txt.

See Also

screen_by_convergence1, screen_by_convergence2, rENM_project_dir

Examples

```
## Not run:
  stage_screened_variables("CASP")
  stage_screened_variables("CASP", top_n = 5, rank_method = "medianPI")

## End(Not run)
```

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